

Tutorial 7 – Numerical Integration Solutions

Prepared for class presentation

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Newton–Cotes formulas, interpolation-based quadrature, composite rules, and area estimation.

- 1 Exercise 1: Newton–Cotes Formulas
- 2 Exercise 2: About Integration
- 3 Exercise 3: Marginal Probability in a Bayesian Network
- 4 Exercise 4: Generalized Trapezoidal and Simpson Methods
- 5 Exercise 5: Robot Sensor Field
- 6 Exercise 6: Kidney-Shaped Swimming Pool
- 7 Final Summary

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The continuous least-squares functional is

$$J(\alpha, \beta) = \int_{1/2}^1 \left(\frac{1}{x^2} - (\alpha x + \beta) \right)^2 dx.$$

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The continuous least-squares functional is

$$J(\alpha, \beta) = \int_{1/2}^1 \left(\frac{1}{x^2} - (\alpha x + \beta) \right)^2 dx.$$

The minimizer satisfies

$$\frac{\partial J}{\partial \alpha} = 0, \quad \frac{\partial J}{\partial \beta} = 0.$$

Exercise 1: Normal equations

From the optimality conditions,

$$\int_{1/2}^1 x \left(\frac{1}{x^2} - \alpha x - \beta \right) dx = 0, \quad \int_{1/2}^1 \left(\frac{1}{x^2} - \alpha x - \beta \right) dx = 0.$$

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After computing the integrals, we obtain

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Equivalently,

$$\begin{cases} 7\alpha + 9\beta = 24 \ln 2, \\ 3\alpha + 4\beta = 8. \end{cases}$$

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Solving the system gives

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$$P_1(x) \approx -5.4579x + 6.0934.$$

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Key idea

The least-squares approximation is not obtained by interpolation; it is obtained by orthogonality of the residual to the basis functions 1 and x .

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Therefore,

$$\boxed{I = 1.}$$

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So,

$$\int_{1/2}^1 P_1(x) \, dx = 1.$$

Exercise 1: Trapezoidal Newton–Cotes formula

The trapezoidal formula on $[a, b]$ is

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Here,

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Comparison

Exact value: $I = 1$. The least-squares integral gives 1, while the trapezoidal rule gives 1.25.

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We want to build an interpolation polynomial P_2 in Newton form:

$$P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1).$$

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The function values are

$$f(1) = 1, \quad f\left(\frac{3}{2}\right) = \frac{2}{3}, \quad f(2) = \frac{1}{2}.$$

Exercise 2: Divided differences

First divided differences:

$$f[x_0, x_1] = \frac{\frac{2}{3} - 1}{\frac{3}{2} - 1} = -\frac{2}{3}, \quad f[x_1, x_2] = \frac{\frac{1}{2} - \frac{2}{3}}{2 - \frac{3}{2}} = -\frac{1}{3}.$$

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Second divided difference:

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{-\frac{1}{3} + \frac{2}{3}}{2 - 1} = \frac{1}{3}.$$

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Thus,

$$P_2(x) = 1 - \frac{2}{3}(x - 1) + \frac{1}{3}(x - 1)\left(x - \frac{3}{2}\right).$$

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Equivalently,

$$P_2(x) = \frac{1}{3}x^2 - \frac{3}{2}x + \frac{13}{6}.$$

Exercise 2: Interpolation error

The interpolation error formula gives

$$f(x) - P_2(x) = \frac{f^{(3)}(\xi_x)}{3!} (x-1) \left(x - \frac{3}{2}\right) (x-2), \quad \xi_x \in [1, 2].$$

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Hence,

$$|f(x) - P_2(x)| \leq (x-1)(2-x) \left|x - \frac{3}{2}\right|.$$

Exercise 2: Quadrature formula

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$$J_2(f) = \lambda_0 f(1) + \lambda_1 f\left(\frac{3}{2}\right) + \lambda_2 f(2).$$

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x	$\frac{3}{2}$
x^2	$\frac{7}{3}$

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Thus the weights satisfy

$$\begin{cases} \lambda_0 + \lambda_1 + \lambda_2 = 1, \\ \lambda_0 + \frac{3}{2}\lambda_1 + 2\lambda_2 = \frac{3}{2}, \\ \lambda_0 + \frac{9}{4}\lambda_1 + 4\lambda_2 = \frac{7}{3}. \end{cases}$$

Exercise 2: Weights and rule

Solving the system gives

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Observation

These are Simpson weights on the interval $[1, 2]$ with midpoint $3/2$.

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Also,

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So the rule is exact for degree 3. For $p(x) = x^4$,

$$\int_1^2 x^4 dx = \frac{31}{5}, \quad J_2(x^4) = \frac{149}{24}.$$

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So the rule is exact for degree 3. For $p(x) = x^4$,

$$\int_1^2 x^4 dx = \frac{31}{5}, \quad J_2(x^4) = \frac{149}{24}.$$

Since $149/24 \neq 31/5$,

degree of exactness = 3.

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For higher accuracy, apply the composite Simpson method on smaller subintervals.

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Let

$$f(x) = x \sin(\pi x).$$

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$$P(E = e \mid X = x) = \sin(\pi x), \quad P(X = x) = x,$$

we obtain

$$P(E = e) = \int_0^1 x \sin(\pi x) \, dx.$$

Let

$$f(x) = x \sin(\pi x).$$

We divide $[0, 1]$ into $n = 4$ subintervals:

$$h = \frac{1 - 0}{4} = \frac{1}{4}.$$

Exercise 3: Nodes and values

The nodes are

$$x_0 = 0, \quad x_1 = \frac{1}{4}, \quad x_2 = \frac{1}{2}, \quad x_3 = \frac{3}{4}, \quad x_4 = 1.$$

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The corresponding values are

i	0	1	2	3	4
x_i	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$f(x_i)$	0	$\frac{\sqrt{2}}{8}$	$\frac{1}{2}$	$\frac{3\sqrt{2}}{8}$	0

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These values will be used in both trapezoidal and Simpson formulas.

Exercise 3: Composite trapezoidal rule

The composite trapezoidal formula is

$$T_4 = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)] .$$

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$$T_4 = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)] .$$

Using $h = 1/4$,

$$T_4 = \frac{1}{8} \left[0 + 2 \cdot \frac{\sqrt{2}}{8} + 2 \cdot \frac{1}{2} + 2 \cdot \frac{3\sqrt{2}}{8} + 0 \right] .$$

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Therefore,

$$T_4 = \frac{1 + \sqrt{2}}{8} .$$

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Therefore,

$$T_4 = \frac{1 + \sqrt{2}}{8}.$$

Numerically,

$$T_4 \approx 0.301777.$$

Exercise 3: Composite Simpson rule

Since $n = 4$ is even, Simpson's rule applies.

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$$S_4 = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)].$$

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$$S_4 = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)].$$

Using $h = 1/4$,

$$S_4 = \frac{1}{12} \left[0 + 4 \cdot \frac{\sqrt{2}}{8} + 2 \cdot \frac{1}{2} + 4 \cdot \frac{3\sqrt{2}}{8} + 0 \right].$$

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Therefore,

$$S_4 = \frac{1 + 2\sqrt{2}}{12}.$$

Exercise 3: Composite Simpson rule

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Therefore,

$$S_4 = \frac{1 + 2\sqrt{2}}{12}.$$

Numerically,

$$S_4 \approx 0.319036.$$

Exercise 3: Exact value and comparison

Using integration by parts,

$$\int x \sin(\pi x) \, dx = -\frac{x \cos(\pi x)}{\pi} + \frac{\sin(\pi x)}{\pi^2}.$$

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$$\int x \sin(\pi x) \, dx = -\frac{x \cos(\pi x)}{\pi} + \frac{\sin(\pi x)}{\pi^2}.$$

Hence,

$$I = \int_0^1 x \sin(\pi x) \, dx = \left[-\frac{x \cos(\pi x)}{\pi} + \frac{\sin(\pi x)}{\pi^2} \right]_0^1 = \frac{1}{\pi}.$$

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Numerically,

$$I \approx 0.318310.$$

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Method	Value	Error (approx. $-I$)
T_4	0.301777	-0.016533
S_4	0.319036	0.000726

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Method	Value	Error (approx. $-I$)
T_4	0.301777	-0.016533
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Simpson's rule is much more accurate here.

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Exercise 4: Exact integral

Let

$$I = \int_0^{\pi} \sin x \, dx.$$

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$$I = [-\cos x]_0^{\pi} = -\cos(\pi) + \cos(0).$$

Since $\cos(\pi) = -1$ and $\cos(0) = 1$,

$$I = 2.$$

Exercise 4: Nodes for $h = \pi/4$

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$$h = \frac{\pi}{4},$$

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The nodes are

$$0, \quad \frac{\pi}{4}, \quad \frac{\pi}{2}, \quad \frac{3\pi}{4}, \quad \pi.$$

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$$h = \frac{\pi}{4},$$

we have

$$n = \frac{\pi - 0}{\pi/4} = 4.$$

The nodes are

$$0, \quad \frac{\pi}{4}, \quad \frac{\pi}{2}, \quad \frac{3\pi}{4}, \quad \pi.$$

For $f(x) = \sin x$,

x_i	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$f(x_i)$	0	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{2}}{2}$	0

Exercise 4: Generalized trapezoidal method

The composite trapezoidal formula gives

$$T_4 = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)] .$$

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$$T_4 = \frac{\pi}{4} (1 + \sqrt{2}).$$

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Numerically,

$$T_4 \approx 1.896119.$$

Exercise 4: Generalized Simpson method

Since $n = 4$ is even, Simpson's rule applies:

$$S_4 = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)] .$$

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Thus,

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Therefore,

$$S_4 = \frac{\pi}{6} (1 + 2\sqrt{2}).$$

Numerically,

$$S_4 \approx 2.004560.$$

Exercise 4: Error estimates

For the composite trapezoidal rule,

$$|E_T| \leq \frac{(b-a)^3}{12n^2} \max_{[a,b]} |f''(x)|.$$

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With $n = 4$,

$$|E_T| \leq \frac{\pi^3}{12 \cdot 4^2} = \frac{\pi^3}{192} \approx 0.161491.$$

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For Simpson's rule,

$$|E_S| \leq \frac{(b-a)^5}{180n^4} \max_{[a,b]} |f^{(4)}(x)|.$$

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For Simpson's rule,

$$|E_S| \leq \frac{(b-a)^5}{180n^4} \max_{[a,b]} |f^{(4)}(x)|.$$

Since $f^{(4)}(x) = \sin x$,

$$|E_S| \leq \frac{\pi^5}{180 \cdot 4^4} = \frac{\pi^5}{46080} \approx 0.006641.$$

Exercise 4: Actual errors

The exact value is

$$I = 2.$$

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For the trapezoidal method,

$$E_T = I - T_4 = 2 - \frac{\pi}{4}(1 + \sqrt{2}).$$

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$$E_T \approx 0.103881.$$

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The trapezoidal rule underestimates the integral.

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$$E_T = I - T_4 = 2 - \frac{\pi}{4}(1 + \sqrt{2}).$$

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The trapezoidal rule underestimates the integral.

For Simpson's method,

$$E_S = I - S_4 = 2 - \frac{\pi}{6}(1 + 2\sqrt{2}).$$

Exercise 4: Actual errors

The exact value is

$$I = 2.$$

For the trapezoidal method,

$$E_T = I - T_4 = 2 - \frac{\pi}{4}(1 + \sqrt{2}).$$

Thus,

$$E_T \approx 0.103881.$$

The trapezoidal rule underestimates the integral.

For Simpson's method,

$$E_S = I - S_4 = 2 - \frac{\pi}{6}(1 + 2\sqrt{2}).$$

Thus,

$$E_S \approx -0.004560.$$

Exercise 4: Actual errors

The exact value is

$$I = 2.$$

For the trapezoidal method,

$$E_T = I - T_4 = 2 - \frac{\pi}{4}(1 + \sqrt{2}).$$

Thus,

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The trapezoidal rule underestimates the integral.

For Simpson's method,

$$E_S = I - S_4 = 2 - \frac{\pi}{6}(1 + 2\sqrt{2}).$$

Thus,

$$E_S \approx -0.004560.$$

Simpson's rule slightly overestimates the integral, but is more accurate.

Exercise 4: Choosing n for trapezoidal error $< 5 \times 10^{-4}$

For the trapezoidal rule,

$$|E_T| \leq \frac{\pi^3}{12n^2}.$$

Exercise 4: Choosing n for trapezoidal error $< 5 \times 10^{-4}$

For the trapezoidal rule,

$$|E_T| \leq \frac{\pi^3}{12n^2}.$$

We require

$$\frac{\pi^3}{12n^2} < 5 \times 10^{-4}.$$

Exercise 4: Choosing n for trapezoidal error $< 5 \times 10^{-4}$

For the trapezoidal rule,

$$|E_T| \leq \frac{\pi^3}{12n^2}.$$

We require

$$\frac{\pi^3}{12n^2} < 5 \times 10^{-4}.$$

So

$$n^2 > \frac{\pi^3}{12 \cdot 5 \times 10^{-4}}.$$

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We require

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So

$$n^2 > \frac{\pi^3}{12 \cdot 5 \times 10^{-4}}.$$

Hence,

$$n > \sqrt{\frac{\pi^3}{12 \cdot 5 \times 10^{-4}}} \approx 71.8868.$$

Exercise 4: Choosing n for trapezoidal error $< 5 \times 10^{-4}$

For the trapezoidal rule,

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So

$$n^2 > \frac{\pi^3}{12 \cdot 5 \times 10^{-4}}.$$

Hence,

$$n > \sqrt{\frac{\pi^3}{12 \cdot 5 \times 10^{-4}}} \approx 71.8868.$$

Therefore choose

$n = 72, \quad h = \frac{\pi}{72}.$

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For Simpson's rule,

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For Simpson's rule,

$$|E_S| \leq \frac{\pi^5}{180n^4}.$$

We require

$$\frac{\pi^5}{180n^4} < 5 \times 10^{-4}.$$

So

$$n > \left(\frac{\pi^5}{180 \cdot 5 \times 10^{-4}} \right)^{1/4} \approx 7.6362.$$

Exercise 4: Choosing n for Simpson error $< 5 \times 10^{-4}$

For Simpson's rule,

$$|E_S| \leq \frac{\pi^5}{180n^4}.$$

We require

$$\frac{\pi^5}{180n^4} < 5 \times 10^{-4}.$$

So

$$n > \left(\frac{\pi^5}{180 \cdot 5 \times 10^{-4}} \right)^{1/4} \approx 7.6362.$$

Since Simpson's rule requires n even, choose

$n = 8, \quad h = \frac{\pi}{8}.$

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Exercise 5: Data and method

The robot records detection widths every 2 meters:

i	0	1	2	3	4	5	6
$f(x_i)$	1.8	2.4	2.6	2.3	2.0	1.7	1.5

Exercise 5: Data and method

The robot records detection widths every 2 meters:

i	0	1	2	3	4	5	6
$f(x_i)$	1.8	2.4	2.6	2.3	2.0	1.7	1.5

The spacing is

$$h = 2.$$

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There are 7 points, hence

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$f(x_i)$	1.8	2.4	2.6	2.3	2.0	1.7	1.5

The spacing is

$$h = 2.$$

There are 7 points, hence

$$n = 6.$$

Since $n = 6$ is even, we use the composite Simpson rule.

Exercise 5: Area calculation

Composite Simpson formula:

$$A \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + f_6].$$

Exercise 5: Area calculation

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$$A \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + f_6].$$

Substitute the data:

$$A \approx \frac{2}{3} [1.8 + 4(2.4) + 2(2.6) + 4(2.3) + 2(2.0) + 4(1.7) + 1.5].$$

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Composite Simpson formula:

$$A \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + f_6].$$

Substitute the data:

$$A \approx \frac{2}{3} [1.8 + 4(2.4) + 2(2.6) + 4(2.3) + 2(2.0) + 4(1.7) + 1.5].$$

Simplify:

$$A \approx \frac{2}{3} [1.8 + 9.6 + 5.2 + 9.2 + 4.0 + 6.8 + 1.5].$$

Exercise 5: Area calculation

Composite Simpson formula:

$$A \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + f_6].$$

Substitute the data:

$$A \approx \frac{2}{3} [1.8 + 4(2.4) + 2(2.6) + 4(2.3) + 2(2.0) + 4(1.7) + 1.5].$$

Simplify:

$$A \approx \frac{2}{3} [1.8 + 9.6 + 5.2 + 9.2 + 4.0 + 6.8 + 1.5].$$

Thus,

$$A \approx \frac{2}{3} (38.1) = 25.4.$$

Exercise 5: Area calculation

Composite Simpson formula:

$$A \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + f_6].$$

Substitute the data:

$$A \approx \frac{2}{3} [1.8 + 4(2.4) + 2(2.6) + 4(2.3) + 2(2.0) + 4(1.7) + 1.5].$$

Simplify:

$$A \approx \frac{2}{3} [1.8 + 9.6 + 5.2 + 9.2 + 4.0 + 6.8 + 1.5].$$

Thus,

$$A \approx \frac{2}{3} (38.1) = 25.4.$$

Therefore,

$A \approx 25.4 \text{ m}^2.$

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The interior widths are

6.2, 7.2, 6.8, 5.6, 5.0, 4.8, 4.7.

Exercise 6: Data and method

The interior widths are

6.2, 7.2, 6.8, 5.6, 5.0, 4.8, 4.7.

Since the pool is kidney-shaped, the endpoint widths are 0.

Exercise 6: Data and method

The interior widths are

6.2, 7.2, 6.8, 5.6, 5.0, 4.8, 4.7.

Since the pool is kidney-shaped, the endpoint widths are 0. Thus,

i	0	1	2	3	4	5	6	7	8
f_i	0	6.2	7.2	6.8	5.6	5.0	4.8	4.7	0

Exercise 6: Data and method

The interior widths are

6.2, 7.2, 6.8, 5.6, 5.0, 4.8, 4.7.

Since the pool is kidney-shaped, the endpoint widths are 0. Thus,

i	0	1	2	3	4	5	6	7	8
f_i	0	6.2	7.2	6.8	5.6	5.0	4.8	4.7	0

The spacing is

$$h = 2.$$

Exercise 6: Data and method

The interior widths are

6.2, 7.2, 6.8, 5.6, 5.0, 4.8, 4.7.

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There are 9 points, hence $n = 8$, which is even.

Exercise 6: Area calculation

Using composite Simpson's rule,

$$A \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + 2f_6 + 4f_7 + f_8].$$

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Substitute $h = 2$ and the data:

$$\begin{aligned} A \approx \frac{2}{3} [0 + 4(6.2) + 2(7.2) + 4(6.8) + 2(5.6) \\ + 4(5.0) + 2(4.8) + 4(4.7) + 0]. \end{aligned}$$

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Simplify:

$$A \approx \frac{2}{3} [24.8 + 14.4 + 27.2 + 11.2 + 20.0 + 9.6 + 18.8].$$

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So,

$A \approx 84 \text{ m}^2.$

- 1 Exercise 1: Newton–Cotes Formulas
- 2 Exercise 2: About Integration
- 3 Exercise 3: Marginal Probability in a Bayesian Network
- 4 Exercise 4: Generalized Trapezoidal and Simpson Methods
- 5 Exercise 5: Robot Sensor Field
- 6 Exercise 6: Kidney-Shaped Swimming Pool
- 7 Final Summary

Exercise	Main result
1	Least-squares integral gives 1; trapezoidal gives $5/4$.
2	Simpson-type rule: $\frac{1}{6}f(1) + \frac{2}{3}f(3/2) + \frac{1}{6}f(2)$; degree 3; $\ln 2 \approx 25/36$.
3	$T_4 \approx 0.301777$, $S_4 \approx 0.319036$, exact $1/\pi \approx 0.318310$.
4	For $\int_0^\pi \sin x \, dx$, $T_4 \approx 1.896119$, $S_4 \approx 2.004560$; choose $n = 72$ trapezoidal, $n = 8$ Simpson for tolerance.
5	Robot field area: 25.4 m^2 .
6	Pool area: 84 m^2 .

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Questions?